2.10.1 Sum to infinite terms of a G.P.

The sum of infinite terms of a G.P. is given by $S_\infty = a + ar + ar^2 + ar^3 + \cdots = \frac{a}{1-r}, -1 < r < 1$

Example 2.46 Find the sum of 8 terms of the G.P. $1, -3, 9, -27\ldots$

Solutions Here, the first term $a = 1$, common ratio $r = \frac{-3}{1} = -3 < 1$, Here, $n = 8$.

Sum to n terms of a G.P. is $S_n = \frac{a(r^n - 1)}{r - 1}$ if $r \neq 1$

$$\text{Hence,} \quad S_8 = \frac{1\left((-3)^8 - 1\right)}{(-3) - 1} = \frac{6561 - 1}{-4} = -1640$$

Example 2.47 Find the first term of a G.P. in which $S_6 = 4095$ and $r = 4$.

Solution Common ratio $= 4 > 1$, Sum of first 6 terms $S_6 = 4095$

$$\text{Hence,} \quad S_6 = \frac{a(r^n - 1)}{r - 1} = 4095$$

$$\because \quad r = 4, \frac{a(4^6 - 1)}{4 - 1} = 4095 \Rightarrow a \times \frac{4095}{3} = 4095$$

$$\text{First term} \quad a = 3.$$

Example 2.48 How many terms of the series $1 + 4 + 16 + \cdots$ make the sum 1365?

Solution Let n be the number of terms to be added to get the sum 1365

$$a = 1,\ r = \frac{4}{1} = 4 > 1$$

$$S_n = 1365 \Rightarrow \frac{a(r^n - 1)}{r - 1} = 1365$$

$$\frac{1(4^n - 1)}{4 - 1} = 1365 \text{ so, } (4^n - 1) = 4095$$

$$4^n = 4096 \text{ then } 4^n = 4^6$$

$$n = 6$$

Example 2.49 Find the sum $3 + 1 + \frac{1}{3} + \ldots\infty$

Solution Here $a = 3$, $r = \frac{t_2}{t_1} = \frac{1}{3}$

Sum of infinite terms $S_\infty = \frac{a}{1-r} = \frac{3}{1 - \frac{1}{3}} = \frac{9}{2}$

Example 2.50 Find the rational form of the number $0.6666\ldots$

Solution We can express the number $0.6666\ldots$ as follows

$$0.6666\ldots = 0.6 + 0.06 + 0.006 + 0.0006 + \cdots$$

We now see that numbers $0.6, 0.06, 0.006\ldots$ form a G.P. whose first term $a = 0.6$ and common ratio $r = \frac{0.06}{0.6} = 0.1$. Also $-1 < r = 0.1 < 1$

Using the infinite G.P. formula, we have

$$0.6666\ldots = 0.6 + 0.06 + 0.006 + 0.0006 + \cdots = \frac{0.6}{1 - 0.1} = \frac{0.6}{0.9} = \frac{2}{3}$$

Thus the rational number equivalent of $0.6666\ldots$ is $\frac{2}{3}$

Progress Check

1. Sum to infinite number of terms of a G.P. is ___.
2. For what values of r, does the formula for infinite G.P. valid?

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